

A Multimodal Image–Text Fusion Framework for Automated Skin Lesion Analysis, ABCDE Evaluation, and Interactive Medical Query Support Using a Patient-Centric Conversational System

1. Project Overview

The project aims to assist in **skin lesion cancer assessment** by gathering multimodal information from the patient such as:

- Skin lesion images
- Medical symptoms and history
- Reports such as biopsy or blood test results (e.g., elevated LDH)
- Patient-provided evolution information (increase in size, color change, itching, spreading)

All data is collected through a **chat-based interface**, where patient responses and images are processed continuously.

The system maintains **chat history** to ensure contextual understanding across all steps.

The overall pipeline includes image preprocessing, lesion analysis, multimodal fusion, VQA-based interaction, and final diagnosis assistance.

2. Stage-Wise Workflow

Stage 1: Preprocessing of Datasets

Purpose

Prepare datasets for training all downstream components.

Preprocessing Performed

- Image resizing, normalization
- Removal of noise, hair artifacts (if required)
- Mask–image pair alignment for segmentation datasets
- Text data cleaning for medical descriptions
- Splitting datasets into train/validation/test sets

Input

- Public skin lesion datasets
- Patient medical datasets

- Collected medical reports

Output

- Clean, standardized datasets ready for training SLS, SLRC, HC, and MedProc models.
-

Stage 2: Skin Lesion Segmentation (SLS)

Purpose

Isolate the lesion area from the full skin image.

Input

- Original high-resolution skin lesion image from camera

Preprocessing Required

- Resize image to model resolution
- Normalize pixel values
- Optional: apply contrast enhancement

Operations Performed

- Generate a **binary mask (B/W segmented lesion)**
- Remove surrounding skin regions

- Identify lesion boundaries for downstream analysis

Output

- Segmentation mask
 - Cropped binary lesion region
 - Geometry information (boundaries, contour, pixel area)
-

Stage 3: Skin Lesion ROI Cropping (SLRC)

Purpose

Extract the **exact lesion region in original color**, using the segmentation mask.

Input

- Original skin image
- Segmentation mask from SLS

Operations Performed

- Align mask with original image
- Crop Region of Interest (ROI) while preserving full lesion borders
- Remove non-lesion parts

Output

- High-quality **color lesion ROI**
 - Updated bounding box coordinates
 - Clean image ready for classification
-

Stage 4: Hierarchical Classification + Raw ABCD Extraction (Updated)

Purpose

Classify skin lesions and compute **image-derived ABCD metrics**.

Input

- Color lesion ROI image from SLRC
- Segmentation geometry from SLS

Preprocessing Required

- Image normalization
- Prepare ROI to correct aspect ratio
- Minor noise reduction if required

Operations Performed

1. Hierarchical Lesion Classification

- Cancerous vs non-cancerous classification
- Subcategory classification (melanoma, nevus, etc.)

2. Extract Image-Based ABCD Values

- **A – Asymmetry:** shape imbalance via mask
- **B – Border Irregularity:** edge complexity
- **C – Color Variation:** number and unevenness of color clusters
- **D – Diameter:** maximum lesion span in pixels → converted to mm

Output

- Image-based A, B, C, D values
 - Classification logits & class predictions
 - Intermediate risk indicators
 - Provenance data (confidence, version)
-

Stage 5: MedProc — Medical Information Processing (Updated)

Purpose

Extract structured information from patient-reported symptoms, medical history, and documents.

Input

- Chat responses describing symptoms
- Uploaded biopsy/blood reports
- Prior lesion images (optional)

Preprocessing Required

- Text normalization
- Medical keyword extraction
- Parsing numerical values from reports (e.g., LDH level)

Operations Performed

- Convert patient answers into clinical indicators (itching, spreading, inflammation)
- Derive **E – Evolution** (increase in size, symptoms, timeline)

- Process non-skin-disease symptoms for general physician support
- Prepare structured clinical features

Output

- Structured Evolution (E) value
 - Symptom vectors
 - Parsed medical findings
 - JSON containing reasoning for each extracted feature
-

Stage 6: Image + Text Fusion Layer (IT-Fusion) — Final Aggregation & Decision Layer (Updated)

Purpose

Combine all image and textual evidence to generate **final clinical conclusions**.

Input

- Raw ABCD values from HC
- E value from MedProc
- Classification logits

- SLS & SLRC metadata (mask quality, scale, bounding boxes)
- Patient symptom summaries

Preprocessing Required

- Normalize numeric features
- Validate timelines, scale, and mask consistency
- Check for missing or conflicting information

Operations Performed

1. Multimodal Aggregation

- Combine image-derived ABCD with text-derived E
- Merge classification results with symptom evidence

2. Rule-Based Decision Checks

- Validate sizes, borders, E progression
- Override conflicts or unsafe outputs
- Normalize diameter dimensions

3. Final Score Computation

- Generate final ABCDE values

- Compute calibrated risk score
- Determine final lesion class

4. **JSON Construction**

- Final classification
- Final ABCDE values
- Risk score
- Explanation referencing each contributing stage
- Flags for missing data or clinician review
- Provenance metadata

Output

A **single unified JSON** used in all downstream modules, including VQA, containing:

- Final ABCDE
- Final class & risk
- Explanation
- Stage contributions
- Patient context features

Stage 7: Virtual Question Answering (VQA)

Purpose

Interact with the patient and answer questions based on fused clinical data.

Input

- Unified JSON from IT-Fusion
- Patient queries

Operations Performed

- Determine which part of JSON is relevant for answering
- Query appropriate modules when needed
- Use reinforcement signals to avoid hallucination
- Maintain conversation context

Output

- Accurate, context-aware responses
- Explanations based on clinical reasoning
- Updated chat history

Stage 8: Frontend–Backend and API Layer

(unchanged except formatting)

Purpose

Provide the user interface and backend communication.

Inputs: User actions, uploaded files, messages

Output: Real-time results, visualizations, chat responses

Stage 9: Testing and Deployment

(unchanged except formatting)

Purpose

Ensure reliability and accuracy.

Operations

- Model accuracy testing
 - Integration testing
 - Performance checks
 - Deployment to production environment
-

End-to-End Flow Summary

1. Patient provides image + symptoms via chat
2. SLS → lesion mask
3. SLRC → clean color ROI
4. HC → classification + raw ABCD
5. MedProc → structured E + patient clinical features
6. IT-Fusion → FINAL ABCDE + risk + JSON + explanation
7. VQA → answers patient queries using JSON
8. Frontend displays results
9. System undergoes testing and gets deployed

Dataset

Stage Name	Dataset Name(s)	Trainin g	Testin g	Validatio n	Total
SLS (Skin Lesion Segmentation)	ISIC 2016 – Task 1 Segmentation	900	379	100	1379
SLS – Extended	ISIC 2017 – Task 1 Segmentation	2000	600	300	2900

SLRC (ROI Cropping)	ROI crops generated from ISIC 2016 + ISIC 2017 segmentation outputs	2900	979	400	4279
HC (Level 1 – Benign vs Malignant)	ISIC 2018 Classification Dataset (10,015 images)	7000	2000	1015	10,015
HC – Combined ISIC Stages	ISIC 2019 (25,331 images) + ISIC 2020 (33,126 images)	40,000	10,000	8,457	58,457
HC – Extended Multi-Class Dermatology	ISIC 2016 Task 3 + ISIC 2016 Task 3B + ISIC 2017 Classification + ISIC 2024 (~18k images)	20,000	8,000	2,900	30,900
HC – Large Clinical Dataset	BCN-20000 (20,000 clinical dermatology images)	14,000	4,000	2,000	20,000
HC – Dermoscopic Pigmented Lesions	MiLK-10K (10,015 dermoscopic images)	7000	2000	1015	10,015
MedProc (Medical Text Understanding)	MIMIC-III + MIMIC-IV Clinical Notes	50,000	10,000	5,000	65,000
IT Fusion (Image + Text Fusion Model)	Derm1M (1M images) + HAM10000 + PAD-UFES-20 + ISIC 2016-2024 + BCN-20000 + MIMIC-III/IV clinical text	700,000	200,000	100,000	1,000,000 +
VQA (Virtual Question Answering)	DermaVQA + SkinCancerVQA + SkinQA (clinical) + PathVQA + ISIC-Derived DermaQA	80,000	15,000	10,000	105,000

Models

Stage name	Model name	Paper name	Is code available	Paper link	Paper citation	Use of model	Advantage of model	Disadvantage of model
SLS (Segmentation)	ARCUNet	ARCUNet: Enhancing Skin Lesion Segmentation with Residual Connections and Attention	Yes (NIH/PMC article typically links to code)	https://pubmed.ncbi.nlm.nih.gov/articles/PMCID/920274/		Dermatology-focused U-Net variant with residual and attention modules for skin lesion segmentation.	Designed specifically for skin lesions; strong boundary accuracy and easy to integrate into existing U-Net pipelines.	Not necessarily the absolute top performer compared to newest transformer hybrids; slightly heavier than plain U-Net.
HC (Classification)	EfficientFormerV2	EfficientFormerV2: A Lightweight Vision Transformer for Efficient Image Recognition	Yes (official GitHub)	https://github.com/snnap-research/EfficientFormer		Backbone for benign vs malignant and multi-class lesion classification on dermoscopy/clinical crops.	Very efficient transformer backbone with good accuracy and low VRAM usage; suitable for multi-head HC on 12 GB GPU.	Not dermatology-specific; may slightly trail heavier Swin/ViT backbones on peak accuracy if you can afford them.
MedProc (medical text)	BioBERT	BioBERT: A Pre-trained Biomedical Language Representation Model for Biomedical	Yes (Hugging Face, etc.)	https://spacknlp.org/2020/08/25/sent-biobert-clinical_base_cased.html		Text encoder for biopsy reports, blood test summaries, and medical descriptions.	Strong performance on biomedical NER and classification; widely used and supported.	Optimized mainly for literature-style biomedical text; not as strong on long, messy EHR notes as newer

		I Text Mining						clinical models.
IT Fusion (handling missing modalities)	MARIA-style transformer	A Multimodal Transformer Model for Incomplete Healthcare Data (MARIA)	Likely yes (arXiv)	https://arxiv.org/abs/2412.14810		Fusion transformer that combines multiple modalities (image, labs, text, structured data) via masked attention to handle missing modalities.	Naturally handles incomplete modality sets, matching your real-world scenario where not all patients have all tests.	Requires custom integration of your encoders; not dermatology-specific; more engineering effort than simple concatenation.
VQA (baseline VLM)	R-LLaVA	R-LLaVA: Improving Med-VQA Understanding Through Visual Region of Interest Guidance	Yes (OpenReview/GitHub)	https://openreview.net/forum?id=HqdlX8yZJP		Med-VQA model that uses ROI information to guide visual attention and improve answers.	Aligns perfectly with your SLRC stage by using lesion ROIs; outperforms standard LLaVA-Med on multiple Med-VQA benchmarks.	General Med-VQA, not dermatology-specific; needs domain-specific fine-tuning and potentially model-size tuning for 12 GB GPU.
SLRC (Skin Lesion ROI Cropping)	OpenCV							
Frontend-Backend and API	React - Frontend , FastAPI-linking							

Literature Survey

Author Name	Year	Paper Title	Link	Difference from NutriDermAI	Insights Gained	Citations
Pan et al.	2025	A Multimodal Vision Foundation Model for Clinical Dermatology (PanDerm)	https://www.nature.com/articles/s41591-025-03747-y	Large dermatology foundation model across multiple imaging modalities; does not implement full conversational assistant, ABCDE workflow, or RL-based VQA like NutriDermAI.	Shows how self-supervised masked latent modeling and CLIP-style alignment on millions of dermatology images yields universal encoders reusable for SLS, HC, and IT Fusion.	Pan et al. (2025). A Multimodal Vision Foundation Model for Clinical Dermatology (PanDerm)

					rm) [https://www.w. nature.com/articles/s41591-025-03747-y]
Zeng et al.	2 0 2 5	MM-Skin: Enhancing Dermatology Vision-Language Model with Expert-Curated Multimodal Dataset	https://arxiv.org/html/2505.06152v1	Builds dermatology VLM on expert-captioned multimodal data; lacks explicit MedProc rule engine, ABCDE metrics, and end-to-end chat deployment.	Demonstrates that expert-written long-form descriptions and multi-modality inputs significantly improve image-text understanding, guiding IT Fusion training and prompting strategies.

						Language Model with Expert-Curated Multimodal Dataset [https://arxiv.org/html/2505.06152v1]
Yan et al.	2025	Derm1M: A Million-Scale Vision-Language Dataset Aligned with Hierarchical Dermatology Ontology	https://arxiv.org/abs/2503.14911	Provides 1M+ image-text pairs focused on pretraining; does not implement a staged assistant with SLS, ABCDE, and RL-based VQA.	Enables robust dermatology vision-language pretraining where DermLIP outperforms biomedical foundations in zero-shot classification, strongly informing IT Fusion backbones.	Yan et al. (2025). Derm1M: A Million-Scale Visi

						on-Lan gua ge Dat ase t Alig ned wit h Hie rarc hic al Der ma tolo gy Ont olo gy [htt ps: //ar xiv. org /ab s/2 503 .14 911]
Aboulmira et al.	2025	Attention-Guided Multimodal Skin Disease Classification	https://www.sciencedirect.com/science/article/pii/S2352914825000887	Combines dermoscopic images and text in an attention-guided classifier; does not include conversational VQA, MedProc reasoning, or multi-stage ISIC/BCN integration.	Shows how attention-based fusion of image and text can improve diagnosis, suggesting architectural patterns for NutriDermAI's multimodal IT Fusion stage.	Aboulmira et al. (2025). · Attention-Guided

					Multimodal Skin Disease Classification [https://www.sciencedirect.com/science/article/pii/S2352914825000887]
Naseem et al.	2024	BioMedBLIP: Advancing Accuracy in Multimodal Medical Tasks Through Biomedically-Informed BLIP Models	https://medinform.jmir.org/2024/1/e56627	Targets radiology and other modalities; not specialized for dermatology or ABCDE-driven workflows.	Demonstrates that domain-adapted BLIP models outperform generic ones on medical tasks, informing model choice and pretraining
					Naseem et al. (2024). BioMedBLIP

					strategy for IT Fusion and VQA.	IP: Advancing Accuracy in Multimodal Medical Tasks Through Biomedically-Informed BLIP Models [https://medinform.jmir.org/2024/1/e56627]
--	--	--	--	--	---------------------------------	---

ScaleFusionNet authors	2025	Transformer-Guided Multi-Scale Feature Fusion for Skin Lesion Segmentation	https://arxiv.org/html/2503.03327v3	Focuses on high-accuracy segmentation using Swin Transformers and deformable convolutions; does not implement ROI color reconstruction, ABCDE scoring, or multimodal reasoning.	Shows multi-scale fusion with Transformers significantly boosts Dice scores on ISIC, making such architectures strong SLS candidates feeding SLRC and downstream modules.	ScaleFusionNet authors (2025). Transformer-Guided Multi-Scale Feature Fusion for Skin Lesion Segmentation [https://arxiv.org/html/2503.03327v3]
------------------------	------	--	---	---	---	---

						7v3]
ESC-UNet authors	2025	ESC-UNET: A Hybrid CNN and Swin Transformers for Skin Lesion Segmentation	https://www.sciencedirect.com/science/article/pii/S2666521225000614	Hybrid CNN–Swin model restricted to segmentation; not embedded in a full conversational assistant or VQA pipeline.	Confirms that combining CNN backbones with Swin Transformers yields high performance on ISIC data, supporting similar hybrid choices for NutriDermAI’s SLS.	ESC-UNet authors (2025). ESC-UNET: A Hybrid CNN and Swin Transformers for Skin Lesion Segmentation [https://www.sciencedirect.co

					m/sci enc e/a rticl e/p ii/S 266 652 122 500 061 4]
MRP-UNet authors	2025	Skin Lesion Segmentation with a Multiscale Input Fusion U-Net	https://www.nature.com/articles/s41598-025-92447-1	Improves segmentation boundaries via multiscale input fusion; does not support ROI reconstruction or text fusion as required in SLRC and IT Fusion.	Demonstrates that multiscale fusion and residual attention refine lesion borders and small-lesion sensitivity, directly benefiting SLS and ABCDE A/B estimation. MR P-U Net aut hor s (20 25) . Ski n Les ion Seg me nta tion wit h a Mul tisc ale Inp ut Fus ion U-N et [htt ps: //w ww. nat

						ure. co m/ arti cle s/s 415 98- 025 -92 447 -1]
Meta-UNet authors	2 0 2 5	Meta-UNet: Enhancing Skin-Lesion Segmentation with Multimodal Metadata	https://pmc.ncbi.nlm.nih.gov/articles/PMC12476315/	Uses metadata to refine segmentation masks; not connected to VQA, ABCDE rule logic, or conversational pipeline.	Shows that including metadata such as lesion location and patient info improves segmentation, encouraging early fusion of structured data in SLS and IT Fusion.	Me ta- UN et aut hor s (20 25) . Me ta- UN et: Enh anc ing Ski n-L esi on Seg me nta tion wit h Mul tim oda l Me tad ata

					[https://pmc.ncbi.nlm.nih.gov/articles/PMC12476315/]
BiADATU-Net authors	2024	Intelligent Skin Lesion Segmentation Using Deformable Attention Transformer U-Net	https://onlinelibrary.wiley.com/doi/10.1111/srt.13783	Focuses purely on deformable-attention-based segmentation; does not integrate text or MedProc reasoning.	Highlights that deformable attention captures irregular lesion borders, improving boundary precision and supporting robust ABCDE border feature extraction.

					on Tra nsf or me r U-N et [htt ps: //o nlin elib rary .wil ey. co m/ doi /10 .11 11/ srt. 137 83]
MUCM-Net authors	2 0 2 4	MUCM-Net: A Mamba Powered UCM-Net for Skin Lesion Segmentation	https://www.explorationpub.com/Journals/em/Article/1001250	Provides lightweight segmentation; does not embed in a larger multimodal, conversational framework.	Delivers an efficient segmentation backbone suitable for 12GB GPUs, aligning with NutriDermAI's hardware constraints for the SLS stage. MU CM -Ne t aut hor s (20 24) . MU CM -Ne t: A Ma mb a Po wer ed UC

						M-Net for Skin Lesion Segmentation [https://www.explorationpub.com/Journals/e/Article/1001250]
Hybrid dual-encoder authors	2024	A Hybrid Deep Learning Approach for Skin Lesion Segmentation Using Dual CNN-ViT Encoders	https://ieeexplore.ieee.org/iel8/6287639/10820123/10910106.pdf	Uses ResNet + ViT dual encoders for segmentation only, without downstream ROI reconstruction or multimodal heads.	Shows that dual-encoder CNN-ViT architectures with attention fusion improve segmentation, offering reusable patterns for SLS and feature extraction feeding IT Fusion.	Hybrid dual-encoder authors (2024). A Hybrid Deep

					Learning Approach for Skin Lesion Segmentation Using Dual CNN-ViT Encoders [https://ieeexplore.ieee.org/iel8/6287639/10820123/10910106.pdf]
--	--	--	--	--	---

Faggioli et al.	2025	Multimodal Learning for Skin Lesion Segmentation and Closed-Ended Visual Question Answering	https://www.dei.unipd.it/~faggioli/temp/clef2025/paper_197.pdf	Jointly tackles segmentation and closed-ended VQA for MEDIQA-MAGIC; does not implement full HC, MedProc, or dialogue management.	Confirms that shared representations between SLS and VQA benefit performance, supporting tight integration of SLS/SLRC with VQA in NutriDermAI.	Faggioli et al. (2025). Multimodal Learning for Skin Lesion Segmentation and Closed-Ended Visual Question Answering [https://www.dei.unipd.it/~faggioli
-----------------	------	---	---	--	---	---

						i/te mp /cle f20 25/ pap er_ 197 .pd f]
IReL team	2025	Tackling Multimodal Dermatology with CLIPSeg-Based Vision-Language Fusion	https://eur-ws.org/Vol-4038/paper_208.pdf	Uses CLIPSeg-based segmentation and multimodal alignment for challenge tasks; lacks ABCDE scoring, rule-based MedProc, and RL answer selection.	Shows CLIP-style joint image-text representations help dermatology segmentation and VQA, informing CLIP-like encoders for IT Fusion and SLS.	IReL team (2025). Tackling Multimodal Dermatology with CLIPSeg-Based Vision-Language Fusion [https://eur-

						ws. org /Vo l-40 38/ pap er_ 208 .pd f]
Review authors	2 0 2 4	A Review of U-Net-Based Deep Learning Models for Skin Lesion Segmentation	https://onlinelibrary.wiley.com/doi/10.1111/ima.70107	Survey of U-Net variants; does not implement a new pipeline or multimodal assistant.	Summarizes segmentation architectures and performance on standard datasets, guiding initial SLS backbone and baseline selection.	Rev iew aut hor s (20 24) . A Rev iew of U-N et- Bas ed Dee p Lea rnin g Mo del s for Ski n Les ion Seg me nta tion [htt ps: //o nlin

					elibrary. com/doi/10.1111/ima.70107]
Multimodal Skin Cancer Prediction authors	2025	Multimodal Skin Cancer Prediction: Integrating Dermoscopic Images and Multi-Modal Data	https://openbioinformaticsjournal.com/VOLUME/18/ELOCATOR/e18750362358444/	Integrates multiple data modalities for prediction but does not extend to conversational interaction, ABCDE rule logic, or VQA.	Shows benefits of fusing dermoscopic images with clinical data, supporting NutriDermAI's plan to combine HC outputs with MedProc in IT Fusion. Multimodal Skin Cancer Prediction: Integrating

					ting Der mo sco pic lma ges and Mul ti-M oda l Dat a [htt ps: //o pen bioi nfo rm atic sjo urn al.c om /V OL UM E/1 8/E LO CA TO R/e 187 503 623 584 44/]
--	--	--	--	--	--

Jütte et al.	2024	Integrating Generative AI with ABCDE Rule Analysis for Early Skin Cancer Detection	https://www.frontiersin.org/journals/medicine/articles/10.3389/med.2024.1445318/full	Uses generative AI plus ABCDE analysis for detection without building a multi-stage assistant with VQA, MedProc, and deployment stack.	Provides a working example of ABCDE-aware generative models for skin cancer, validating NutriDermAI's ABCDE-centric risk analysis design.	Jütte et al. (2024). Integrating Generative AI with ABCDE Rule Analysis for Early Skin Cancer Detection [https://www.frontiersin.org/journals/
--------------	------	--	---	--	---	--

						me dici ne/ arti cle s/1 0.3 389 /fm ed. 202 4.1 445 318 /ful l]
Chatterjee et al.	2021	Dermatological Expert System Implementing the ABCD Rule for Melanoma Detection	https://www.sciencedirect.com/science/article/abs/pii/S0957417420309337	Implements rule-based ABCD expert system using classical processing, not deep segmentation, multimodal fusion, or conversational flow.	Offers formalization of ABCD criteria and total dermoscopic scores as rules, directly informing MedProc's rule-based component.	Ch att erje e et al. (20 21) . Der ma tolo gic al Exp ert Sys tem Imp lem enti ng the AB CD Rul e for Mel ano

					ma Det ecti on [htt ps: //w ww. sci enc edir ect. co m/ sci enc e/a rticl e/a bs/ pii/ S09 574 174 203 093 37]
Senan et al.	2 0 2 1	Analysis of Dermoscopy Images by Using ABCD Rule for Melanoma Diagnosis	https://www.sci encedirect.com/ science/article/ pii/S2666285X2 1000017	Applies ABCD rule with classical methods; no VQA, LLMs, or multi-stage integration.	Reinforces clinical validity of ABCD rule and shows practical algorithms for automatic ABCD evaluation feeding HC and MedProc. Sen an et al. (20 21) · An aly sis of Der mo sco py Ima ges by Usi

					ng AB CD Rul e for Mel ano ma Dia gno sis [htt ps: //w ww. sci enc edir ect. co m/ sci enc e/a rticl e/p ii/S 266 628 5X 210 000 17]
Tas et al.	2 0 2 2	Faster Evaluation of ABCD Rule and Total Dermoscopic Score for Nevomelanocyt ic Lesions	https://brieflands.com/journals/ijssc/articles/102138	Optimizes ABCD/TDS computation speed; does not integrate into a multi-module conversational pipeline.	Demonstrates that ABCD/TDS evaluation can be accelerated for near real-time use, supporting responsive ABCDE scoring in interactive sessions. Tas et al. (20 22) · Fas ter Eva lua tion of

					AB CD Rule and Total Der mo sco pic Score for Nev om ela noc ytic Les ion s [htt ps: //br iefl and s.c om /jo urn als/ jss c/a rticl es/ 102 138]
--	--	--	--	--	---

Shetty et al.	2022	Skin Lesion Classification of Dermoscopic Images Using Hybrid CNN Models and ABCD Rule Features	https://www.nature.com/articles/s41598-022-22644-9	Combines CNN features with ABCD descriptors for classification but omits text reasoning, MedProc, and conversational frontends.	Validates the benefit of explicitly incorporating ABCD-based features into deep classifiers, informing HC feature engineering.	Shetty et al. (2022). Skin Lesion Classification of Dermoscopic Images Using Hybrid CNN Models and ABCD Rule Features [https://www.nature.co
---------------	------	---	---	---	--	--

					m/ arti cle s/s 415 98- 022 -22 644 -9]
Mevorach et al.	2025	A Comparison of Skin Lesions' Diagnoses Between AI-Based Systems and Dermatologists Using ABCDE Rule	https://pmc.ncbi.nlm.nih.gov/articles/PMC12071753/	Evaluates existing AI systems vs dermatologists on ABCDE-based diagnosis; does not propose a new architecture.	Provides clinical evidence that ABCDE-based AI assessment can approach dermatologist performance, supporting NutriDermAI's ABCDE emphasis.

					Using ABCDE Rule [https://pmc.ncbi.nlm.nih.gov/articles/PMC12071753/]
Yim et al.	2024	DermaVQA: A Multilingual Visual Question Answering Dataset for Dermatology	https://dl.acm.org/doi/10.1007/978-3-031-72086-4_20	Releases dermatology VQA dataset and baselines but not a full multi-stage assistant with ABCDE, MedProc, and RL-based VQA.	Defines dermatology-specific VQA tasks and multilingual questions, guiding VQA training targets and evaluation for NutriDermAI.

						ing Dat ase t for Der ma tolo gy [htt ps: //dl .ac m.o rg/ doi /10 .10 07/ 978 -3-0 31- 720 86- 4_2 0]
Saeed et al.	2024	MediFact at MEDIQA-M3G 2024: Medical Question Answering with Weakly Supervised Answer Generation	https://aclanthology.org/2024.clinalnlp-1.31/	Uses weak supervision for answer generation in multimodal medical tasks; lacks explicit ABCDE computation and full lesion pipeline.	Shows how weakly supervised and knowledge-aug mented training can compensate for scarce labels in medical VQA, applicable to NutriDermAI's training regime.	Sae ed et al. (20 24) . Me diF act at ME DIQ A- M3 G 202 4: Me dic

					al Qu esti on Ans wer ing wit h We akl y Sup ervi sed Ans wer Ge ner atio n [htt ps: //a cla nth olo gy. org /20 24. clin ical nlp- 1.3 1/]
--	--	--	--	--	--

Kim et al.	2024	TEAM MIPAL at MEDIQA-M3G 2024: Large VQA Models for Multimodal Medical Answer Generation	https://aclanthology.org/2024.clinalnlp-1.30.pdf	Adapts large VQA models for dermatology tasks but does not decompose pipeline into SLS, HC, MedProc, or use RL answer selection.	Provides fine-tuning and answer ranking strategies for large multimodal VQA, informing prompt and scoring design for NutriDermAI's VQA stage.	Kim et al. (2024). TEAM MIPAL at MEDIQA-M3G 2024: Large VQA Models for Multimodal Medical Answer Generation [https://aclanthology.org/2024.clinalnlp-1.30.pdf]
------------	------	--	---	--	---	--

						gy. org /20 24. clin ical nlp- 1.3 0.p df]
H3N1 Lab	2025	DermKEM: A Knowledge-Enhanced Ensemble for Dermatology VQA	https://eur-ws.org/Vol-4038/paper_202.pdf	Ensembles VLMs with external knowledge; does not tie in ABCDE scoring or structured MedProc outputs as knowledge sources.	Shows that knowledge-enhanced ensembles improve VQA accuracy, suggesting integration of MedProc's JSON reasoning into VQA as structured knowledge.	H3 N1 Lab (20 25) . Der mK EM: A Kn owl edg e-E nha nce d Ens em ble for Der ma tolo gy VQ A [htt ps: //c eur- ws. org /Vo l-40 38/

						paper_202.pdf]
Madavan et al.	2025	Med-GRIM: Enhanced Zero-Shot Medical VQA Using Prompt Ensembles	https://openaccess.thecvf.com/content/ICCV2025W/MRR%202025/papers/Madavan_Med-GRIM_Enhanced_Zero-Shot_Medical_VQA_using_prompt-e.pdf	Targets zero-shot Med-VQA using prompt ensembles; not dermatology-specific and lacks RL-based training.	Demonstrates that combining prompts and VLMs improves zero-shot performance, inspiring ensemble and reranking strategies for NutriDermAI's VQA.	Madavan et al. (2025). Med-GRIM: Enhanced Zero-Shot Medical VQA Using Prompt Ensembles [https://openaccess.thecvf.co

					m/ con ten t/IC CV 202 5W /M RR %2 020 25/ pap ers /M ada van _M ed- GRI M_ Enh anc ed_ Zer o-S hot _M edi cal _V QA _us ing _pr om pt-e .pd f]
--	--	--	--	--	---

Li et al.	2022	Medical Visual Question Answering via Corresponding Feature Fusion and Semantic Attention	https://www.aimspress.com/article/doi/10.3934/mbe.2022478	Designs semantic attention and feature fusion for Med-VQA on radiology, not dermatology.	Introduces corresponding feature fusion between image and question with semantic attention, useful for designing NutriDermAI's IT Fusion/VQA heads.	Li et al. (2022). Medical Visual Question Answering via Corresponding Feature Fusion and Semantic Attention [https://www.aimspress.com/article/
-----------	------	---	---	--	---	---

						doi /10 .39 34/ mb e.2 022 478]
DSAP authors	2024	Bridging the Semantic Gap in Medical Visual Question Answering with Dynamic Semantic-Adaptive Prompting	https://ieeexplore.ieee.org/iel8/42/11230061/11039185.pdf	Uses adaptive prompting to improve Med-VQA; not tailored to dermatology or ABCDE.	Shows dynamic prompts can narrow the semantic gap between image and question, suggesting a prompt-adaptor layer for NutriDermAI's VQA.	DSAP authors (2024). Bridging the Semantic Gap in Medical Visual Question Answering with Dynamic Semantic-Ad

					https://ieeexplore.ieee.org/iel8/42/1230061/11039185.pdf
MedVLM-R1 authors	2025 0 Medical Reasoning 5 Capability with GRPO for VQA	https://papers.miccai.org/miccai-2025/paper/3267_paper.pdf	Applies GRPO RL to radiology VQA; not specifically for dermatology or ABCDE scoring.	Provides an RL framework with format and accuracy rewards for VQA, directly informing RL-based training for dermatology VQA in NutriDermAI.	MedVLM-R1 authors (2025). MedVLM-R1: Incentivizing Medical Reason

						ing Ca pab ility wit h GR PO for VQ A [htt ps: //p ape rs. mic cai. org /mi cca i-20 25/ pap er/ 326 7_p ape r.pd f]
RL VQA Validation authors	2 0 2 3	A Reinforcement Learning Approach for VQA Validation	https://www.sciencedirect.com/science/article/pii/S136184152300083X	Uses RL to validate and improve generic VQA answers; not focused on medical images.	Shows RL can select or filter better answers, supporting NutriDermAI's use of RL to reduce hallucinations and improve reliability.	RL VQ A Vali dati on aut hor s (20 23) . A Rei nfo rce me

					nt Lea rnin g Ap pro ach for VQ A Vali dati on [htt ps: //w ww. sci enc edir ect. co m/ sci enc e/a rticl e/p ii/S 136 184 152 300 083 X]
MedThink authors	2 0 2 5	MedThink: A Rationale-Guide d Framework for Explaining Medical Visual Question Answering	https://aclanthology.org/2025.findings-naacl.415.pdf	Creates rationale-genera ting Med-VQA models; not dermatology-spe cific or ABCDE-aware.	Demonstrates how to generate textual rationales alongside answers, aligning with NutriDermAI's goal of JSON explanations and Me dTh ink aut hor s (20 25) . Me dTh

					interpretable responses.	ink: A Ration ale-Gui ded Fra me wor k for Exp lain ing Me dic al Vis ual Qu esti on Ans wer ing [htt ps: //a cla nth olo gy. org /20 25. find ing s-n aac l.41 5.p df]
--	--	--	--	--	--------------------------	---

HEALMQA authors	2024	HEALMQA: A Healthcare Multimodal Question Answering Benchmark	https://openreview.net/pdf/1d0a6992d0bc72959294fe86ec6553f6cbd3ee7a.pdf	Provides a broad healthcare multimodal QA benchmark; not dermatology or ABCDE-focused.	Offers benchmark design and evaluation ideas for multimodal QA, useful to structure NutriDermAI's test scenarios.	HEALMQA authors (2024). HEALMQA: A Healthcare Multimodal Question Answering Benchmark [https://openreview.net/pdf/1d0a6992d0bc7
-----------------	------	---	---	--	---	--

					295 929 4fe 86e c65 53f 6cb d3e e7a .pd f]
Lozano et al.	2 0 2 5	Synthesizing High-Quality Visual Question Answering from Medical Literature	https://arxiv.org/ html/2510.2586 7	Generates VQA samples from text; not focused on dermatology images or ABCDE.	Shows how to synthesize medically grounded VQA data, which can be adapted to generate extra dermatology QA pairs for training. Loz ano et al. (20 25) . Syn the sizi ng Hig h-Q uali ty Vis ual Qu esti on Ans wer ing fro m Me dic al Lite rat ure [htt ps: //ar xiv.

						org /ht ml/ 251 0.2 586 7]
Sviridov et al.	2025	Medical Multimodal Multi-agent Dialogue Benchmark	https://aclanthology.org/2025.emnlp-main.1353.pdf	Designs a multi-agent medical dialogue benchmark; not specific to dermatology or ABCDE rules.	Highlights multi-agent setups and metrics for diagnostic dialogue, offering patterns for NutriDermAI's dialogue and escalation design.	Sviridov et al. (2025). Medical Multimodal Multi-agent Dialogue Benchmark [https://aclanthology.org/2025.emnlp-main.1

						353 .pdf]
Dermatology Chatbot authors	2024	Dermatology Chatbot: An AI-Driven Solution for Accessible Skin Care	https://www.ama-assn.org/publications/journals/dermatology/article/2024/01/01/dermatology-chatbot-an-ai-driven-solution-for-accessible-skin-care	Implements an image+text dermatology chatbot with Hybrid U-Net and MobileNet-V3 but without multimodal LLMs, ABCDE rules, or RL-based VQA.	Shows real-world implementation of dermatology chatbot integrating CNNs and rule-based logic, validating NutriDermAI's multi-stage assistant concept.	Der ma tolo gy Ch atb ot aut hor s (20 24) . Der ma tolo gy Ch atb ot: An AI- Driv en Sol utio n for Acc ess ible Ski n Car e [htt ps: //w ww. am eric asp g.c

					om /art icle /pd f/3 773]
Case study authors	2 0 2 4	Design Considerations from a Wizard-of-Oz Dermatology Case Study with a Conversational Agent	https://dl.acm.org/doi/10.1145/3613905.3651891	Explores semi-human-in-t he-loop dermatology conversational interface; not fully automated or multi-stage like NutriDermAI.	Identifies UX and safety considerations (e.g., disclaimers, escalation) for dermatology chatbots, guiding frontend and interaction design. Cas e stu dy aut hor s (20 24) . Des ign Co nsi der atio ns fro m a Wiz ard-of- Oz Der ma tolo gy Cas e Stu dy wit h a Co nve rsa tion al Ag

					ent [htt ps: //dl .ac m.o rg/ doi /10 .11 45/ 361 390 5.3 651 891]
AMIE authors	2 0 2 5	Towards Conversational Diagnostic Artificial Intelligence (AMIE)	https://www.nature.com/articles/s41586-025-08866-7	Provides a general-purpose conversational diagnostic LLM; not dermatology- or ABCDE-specific.	Gives evaluation framework comparing AI vs clinicians and safety practices, which NutriDermAI can adapt for validation and deployment. AM IE aut hor s (20 25) . To war ds Co nve rsa tion al Dia gno stic Arti fici al Inte llig enc e (A MIE)

					[https://www.nature.com/articles/s41586-025-08866-7]
MIRA framework authors	2025	A Novel Framework for Fusing Modalities in Medical Retrieval-Augmented Generation	https://arxiv.org/html/2507.07902v1	General medical RAG fusion framework; not specifically dermatology or ABCDE-oriented.	Introduces attention-based modality fusion with query-dependent weighting, informing NutriDermAI's IT Fusion and retrieval design.

						reviewal-Augmented Generation [https://arxiv.org/html/2507.07902v1]
Transformer Attention Fusion authors	2025	Transformer Attention Fusion for Fine-Grained Medical Image Analysis	https://pmc.ncbi.nlm.nih.gov/articles/PMC12216456/	Applies dual attention mechanisms to other medical images (e.g., retinopathy); not dermatology-focused.	Shows that combining self- and cross-attention at multiple scales improves fine-grained recognition, relevant for detailed lesion representation.	Transformer Attention Fusion authors (2025). Transformer Attention

					Fusion for Fine-Grained Medical Image Analysis [https://pubmed.ncbi.nlm.nih.gov/articles/PMC12216456/]
Attention Mechanism review authors	2023	Deep Learning Attention Mechanism in Medical Image Analysis	https://www.scitep.com/journals/ijndi/articles/2504000011	Review of attention mechanisms across medical imaging tasks; no specific dermatology pipeline.	Synthesizes attention patterns that can be plugged into SLS, HC, and IT Fusion modules for NutriDermAI.
					Attention Mechanism review authors (2023)

					. Deep Learning Attention Mechanism in Medical Image Analysis [https://www.sciltp.com/journals/ijndi/articles/25040010011]
--	--	--	--	--	---

Swin Medical authors	2025	A Multi-Scale Attention-Based Swin Transformer Model for Medical Image Segmentation	https://www.nature.com/articles/s41598-025-22649-0	Generic Swin-based segmentation model; not dermatology-specific.	Validates Swin Transformers with multi-scale attention for segmentation, reinforcing their use in NutriDermAI's SLS.	Swin Medical authors (2025). A Multi-Scale Attention-Based Swin Transformer Model for Medical Image Segmentation [https://www.nature.co
----------------------	------	---	---	--	--	---

						m/ arti cle s/s 415 98- 025 -22 649 -0]
ISIC organizers	2 0 1 6 - 2 0 2 4	ISIC 2016–2024 Skin Lesion Segmentation and Classification Datasets	https://challenge.isic-archive.com/	Provide benchmark segmentation and classification datasets, but no multi-stage conversational system.	Supply core training and evaluation data for SLS and HC stages (including ISIC 2024 SLICE-3D extension) consistent with NutriDermAI's dataset plan.	ISI C org ani zer s (20 16- 202 4). ISI C 201 6– 202 4 Ski n Les ion Seg me nta tion and Cla ssif icat ion Dat ase ts [htt ps: //c hall

						eng e.is ic-a rchi ve. co m/]
Tschandl et al.	2018	HAM10000: A Large Collection of Multi-Source Dermatoscopic Images	https://www.nature.com/articles/sdata2018161	Dermatoscopic dataset; does not include pipeline or multimodal reasoning.	Offers 10,015 labeled images across seven lesion types, supporting HC training and evaluation.	Tschandl et al. (2018). HAM10000: A Large Collection of Multi-Source Dermatoscopic Images [https://www.nature.com/article

						s/s dat a20 181 61]
Combalia et al.	2019	BCN20000: Dermoscopic Lesions in the Wild	https://www.nature.com/articles/s41597-024-03387-w	Clinical and dermoscopy dataset; not coupled with VQA or ABCDE logic.	Provides large-scale clinical images that extend HC to real-world clinical photos beyond dermoscopy.	Co mb alia et al. (20 19) . BC N2 000 0: Der mo sco pic Les ion s in the Wil d [htt ps: //w ww. nat ure. co m/ arti cle s/s 415 97- 024 -03 387 -w]

He et al.	2020	PathVQA: 30000+ Questions for Medical Visual Question Answering	https://arxiv.org/abs/2003.10286	Pathology VQA dataset; not dermatology-specific.	Pioneers medical VQA dataset design and evaluation metrics transferable to dermatology VQA design.	He et al. (2020). PathVQA: 30000+ Questions for Medical Visual Question Answering [https://arxiv.org/abs/2003.10286]
-----------	------	---	---	--	--	--

Rajkhowa et al.	2024	TM-PathVQA: 90000+ Textless Multilingual Questions for Pathology VQA	https://www.isc-a-archive.org/interspeech_2024/rajkhowa24_interspeech.pdf	Extends PathVQA to multilingual settings; not focused on skin.	Shows that multi-lingual medical VQA is feasible, suggesting multilingual goals for NutriDermAI.	Rajkhowa et al. (2024). TM-PathVQA: 90000+ Textless Multilingual Questions for Pathology VQA [https://www.isc-a-archive.org/interspeech_2024/rajk
-----------------	------	--	---	--	--	---

						howa24_interspeech.pdf]
Yim et al.	2024	DermaVQA: Multilingual Visual Question Answering Dataset for Dermatology	https://papers.miccai.org/miccai-2024/211-Paper2444.html	Dermatology VQA dataset; does not define a full assistant system.	Directly supplies dermatology VQA data for training and evaluating NutriDermAI's VQA module.	Yim et al. (2024). DermaVQA: Multilingual Visual Question Answering Dataset for Dermatology [https://papers.miccai.org

						/m cca i-20 24/ 211 -Pa per 244 4.ht ml]
Johnson et al.	2016	MIMIC-III Clinical Database v1.4	https://physionet.org/content/mimiciii/1.4/	ICU clinical text dataset; not dermatology-focused.	Provides large-scale clinical notes for pretraining MedProc's NLP models and reasoning components.	Joh nso n et al. (20 16) . MI MI C-II I Clin ical Dat aba se v1. 4 [htt ps: //p hys ion et.o rg/ con ten t/m imi ciii/ 1.4 /]

Hyland et al.	2023	MIMIC-IV-Note: Deidentified Free-Text Clinical Notes	https://physionet.org/content/mimic-iv-note/	Updated ICU text dataset; not specific to skin diseases.	Offers more recent clinical notes to improve general medical reasoning in MedProc.	Hyland et al. (2023). MIMIC-IV-Note: Deidentified Free-Text Clinical Notes [https://physionet.org/content/mimic-iv-note/]
Lau et al.	2018	VQA-RAD: Radiology Visual Question Answering Dataset	https://collab.dvb.bayern/spaces/TUMdlma/pages/73379920/	Radiology VQA dataset rather than dermatology.	Provides early Med-VQA design and question taxonomy useful for structuring dermatology VQA.	Lau et al. (2018). VQA-RAD:

						Rad iolo gy Vis ual Qu esti on Ans wer ing Dat ase t [htt ps: //c olla b.d vb. bay ern /sp ace s/T UM dIm a/p age s/7 337 992 0/]
Lin et al.	2 0 2 3	Medical Visual Question Answering: A Survey	https://www.sciencedirect.com/science/article/abs/pii/S0933365723001252	Survey of Med-VQA; no specific new model.	Summarizes methods and datasets for Med-VQA, informing NutriDermAI's VQA design space.	Lin et al. (20 23) . Me dic al Vis ual Qu esti

					on Ans wer ing: A Sur vey [htt ps: //w ww. sci enc edir ect. co m/ sci enc e/a rticl e/a bs/ pii/ S09 333 657 230 012 52]
Natural Language Processing of MIMIC authors	2 0 1 9	Natural Language Processing of MIMIC-III Clinical Notes	https://arxiv.org/ pdf/1912.12397. pdf	Applies NLP to clinical notes; not dermatology-spe cific.	Shows how to preprocess and model clinical free-text for entity and concept extraction, informing MedProc's pipeline. Nat ural Lan gua ge Pro ces sin g of MI MI C aut hor s

					(2019). Natural Language Processing of MICLIC-III Clinical Notes [https://arxiv.org/pdf/1912.12397.pdf]
Clinical Text Review authors	2020	Clinical Text Data in Machine Learning: Systematic Review	https://pmc.ncbi.nlm.nih.gov/articles/PMC7157505/	Systematic review; no new model.	Outlines best practices and challenges for clinical NLP, guiding MedProc design. Clinical Text Review authors (2020). Clin

					ical Tex t Dat a in Ma chi ne Lea rnin g: Sys te ma tic Rev iew [htt ps: //p mc. ncb i.nl m.n ih.g ov/ arti cle s/P MC 715 750 5/]
ChatGPT Dermatology review authors	2 0 2 5	The Role of ChatGPT in Dermatology Diagnostics	https://pmc.ncbi.nlm.nih.gov/articles/PMC12191462/	Review of LLM usage; not a specific deployed system like NutriDermAI.	Highlights strengths, risks, and limitations of LLMs in dermatology, supporting safe integration strategies. Ch atG PT Der ma tolo gy revi ew aut hor s (20

					25) . The Role of ChatGPT in Dermatology Diagnostics [https://pmc.ncbi.nlm.nih.gov/articles/PMC12191462/]
Conversational Agents Effectiveness authors	2025	Effectiveness of AI-Driven Conversational Agents in Improving Dermatological Patient Engagement	https://www.jmir.org/2025/1/e69639/	Evaluates chat agents; does not propose NutriDermAI-like pipeline.	Provides evidence that conversational agents improve patient engagement, supporting NutriDermAI's chat interface. Conversational Agents Effectiveness

					aut hor s (20 25) . Eff ecti ven ess of AI- Driv en Co nve rsa tion al Ag ent s in Imp rovi ng Der ma tolo gic al Pat ient Eng age me nt [htt ps: //w ww. jmir .org /20 25/ 1/e 696
--	--	--	--	--	--

						39/]
Medical Multimodal LLM review authors	2024	Medical Multimodal Large Language Models: A Systematic Review	https://www.sciencedirect.com/science/article/pii/S2950261625000512	Review of MMLLMs; not dermatology-specific.	Summarizes current multimodal LLM architectures and limitations, helping position NutriDermAI relative to state of the art.	Medical Multimodal LLM review authors (2024). Medical Multimodal Large Language Models: A Systematic Review [https://www.scienc

					edir ect. co m/ sci enc e/a rticl e/p ii/S 295 026 162 500 051 2]
MedVLThinker authors	2 0 2 5	MedVLThinker: Simple Baselines for Multimodal Medical Reasoning	https://arxiv.org/ html/2508.0266 9v2	Proposes simple baselines for multimodal reasoning; not dermatology-foc used.	Shows that carefully designed simple multimodal baselines can perform competitively, suggesting efficient designs suited to 12GB hardware.

					son ing [htt ps: //ar xiv. org /ht ml/ 250 8.0 266 9v2]
Validation Dermatology MLLM authors	2 0 2 5	Validation of a Dermatology-Fo cused Multimodal Large Language Model	https://pmc.ncbi.nlm.nih.gov/articles/PMC12608998/	Validates a dermatology MLLM clinically; does not describe a full multi-stage assistant pipeline.	Demonstrates clinical benefit of domain-specific multimodal models, supporting NutriDermAI's specialization strategy. Vali dati on Der ma tolo gy ML LM aut hor s (20 25) . Vali dati on of a Der ma tolo gy- Foc use d Mul tim oda l Lar ge

						Lan gua ge Mo del [htt ps: //p mc. ncb i.nl m.n ih.g ov/ arti cle s/P MC 126 089 98/]
ImageCLEF MEDIQA-MAGI C organizers	2 0 2 5	ImageCLEFmed MEDIQA-MAGIC: Multimodal Medical Dialogue and Generation Benchmark	https://www.imageclef.org/2025/medical/mediqua	Benchmark challenge; not a specific assistant.	Provides task definitions and metrics for multimodal medical dialogue and QA that NutriDermAI can use for evaluation.	Ima ge CL EF ME DIQ A- MA GIC org ani zer s (20 25) · Ima ge CL EF me d ME DIQ

					A- MA GIC : Mul tim oda l Me dic al Dial ogu e and Ge ner atio n Ben ch ma rk [htt ps: //w ww. ima gec lef. org /20 25/ me dic al/ me diq a]
--	--	--	--	--	--

WoundcareVQA authors	2025	WoundcareVQA: A Multilingual Visual Question Answering Dataset for Wound Care	https://www.sciencedirect.com/science/article/pii/S153204625001170	Wound care VQA dataset; adjacent but not dermatology-specific.	Illustrates the construction of skin-related multilingual VQA datasets, providing patterns transferable to dermatology.	WoundcareVQA authors (2025). WoundcareVQA: A Multilingual Visual Question Answering Dataset for Wound Care [https://www.sciencedirect.co
----------------------	------	---	---	--	---	--

					m/ sci enc e/a rticl e/p ii/S 153 204 642 500 117 0]
Skin Cancer Classification Perspectives authors	2 0 2 4	Decoding Skin Cancer Classification: Perspectives, Insights, and Future Directions	https://www.nature.com/articles/s41598-024-81961-3	Perspectives review; not an implementation of a pipeline.	Provides technical and clinical perspectives on skin cancer AI, informing HC design and evaluation choices. Ski n Ca nce r Cla ssif icat ion Per spe ctiv es aut hor s (20 24) . Dec odi ng Ski n Ca nce r Cla ssif icat ion: Per spe

					ctiv es, Insi ght s, and Fut ure Dir ecti ons [htt ps: //w ww. nat ure. co m/ arti cle s/s 415 98- 024 -81 961 -3]
--	--	--	--	--	---